

Cold Wave Survival Guide — Hypothermia and Frostbite First Aid

This guide covers how to survive extreme cold, recognise hypothermia and frostbite, and treat cold-related emergencies when medical help is not immediately available. Based on American Red Cross, Mayo Clinic, and CDC guidelines. In India, cold waves affect Jammu and Kashmir, Himachal Pradesh, Uttarakhand, the Indo-Gangetic plain (Delhi, UP, Bihar), and Rajasthan during winter months.

HYPOTHERMIA CAN KILL AT TEMPERATURES ABOVE 10°C (50°F) when a person is wet, in wind, or exhausted. You do not need freezing temperatures to develop life-threatening hypothermia. Rain, wind, and physical exhaustion are enough.

SECTION 1: HYPOTHERMIA — RECOGNISE THE STAGES

Hypothermia occurs when the body's core temperature drops below 35°C (95°F). It develops in stages and becomes rapidly life-threatening.

Mild hypothermia (core temperature 32–35°C):

- Intense, uncontrollable shivering
- Numbness in hands and feet
- Difficulty with fine motor tasks (fumbling with buttons, dropping things)
- Slurred speech
- Mild confusion or poor judgement

Shivering is the body's attempt to generate heat. It is a warning sign, not a harmless response. A person who is shivering severely needs to be warmed immediately.

Moderate hypothermia (core temperature 28–32°C):

- Shivering stops — this is a DANGEROUS sign, not improvement. It means the body can no longer generate heat.
- Severe confusion, irrational behaviour (may remove clothing in cold — "paradoxical undressing")
- Extreme drowsiness, difficulty staying awake
- Muscle stiffness and loss of coordination
- Slow, weak pulse

Severe hypothermia (core temperature below 28°C):

- Unconsciousness
- Very slow or absent breathing
- No detectable pulse
- The person may appear dead

NEVER assume a cold, unresponsive person is dead. Hypothermia victims have been resuscitated after appearing lifeless. Begin CPR and continue until medical professionals take over. The saying in emergency medicine is: "A person is not dead until they are warm and dead."

SECTION 2: FIRST AID FOR HYPOTHERMIA — STEP BY STEP

Step 1. Call emergency services (112) for any person with moderate or severe hypothermia.

Step 2. Move the person to a warm, sheltered place. Get them out of wind, rain, and cold.

Step 3. Handle the person GENTLY. Do not jostle, shake, or rub the person vigorously. A hypothermic heart is electrically unstable and can go into cardiac arrest from rough handling.

Step 4. Remove ALL wet clothing. Wet clothing conducts heat away from the body 25 times faster than air. Replace with dry clothing or wrap in dry blankets.

Step 5. Warm the CORE first — the chest, abdomen, armpits, and neck. Do NOT warm the arms and legs first. Warming the extremities first causes cold blood from the limbs to rush to the heart, which can trigger cardiac arrest (called "afterdrop").

Step 6. Use gradual warming methods: dry blankets, a sleeping bag, or body-to-body warming (another person sharing a sleeping bag or blankets). Electric blankets or heating pads are acceptable if wrapped in a towel first — never place them directly on skin.

Step 7. If the person is conscious and able to swallow without difficulty, give warm (not hot) sweet drinks — warm water, sweet tea, or broth. **Do NOT give alcohol** — it causes further heat loss by dilating blood vessels.

Step 8. If the person is unconscious and not breathing, begin CPR and continue until emergency services arrive.

DO NOT: Rub the limbs vigorously to "warm up" the person. Do not place in a hot bath. Do not give alcohol. Do not warm hands and feet before warming the core. These actions can all cause cardiac arrest in a hypothermic patient.

SECTION 3: FROSTBITE — RECOGNISE AND TREAT

Frostbite occurs when skin and underlying tissue freeze. It most commonly affects fingers, toes, ears, nose, and cheeks. It can occur alongside hypothermia — always check for both.

Stages of frostbite:

Frostnip (very early frostbite): Skin turns red. Pins-and-needles sensation or throbbing. Skin is still soft. No permanent damage if treated now.

Superficial frostbite: Skin turns pale, waxy, or grayish. Skin feels hard on the surface but soft underneath. The area is numb. Blisters may develop 12 to 36 hours after rewarming.

Deep frostbite: Skin is completely hard, cold, and white or black. The entire thickness of skin and sometimes underlying tissue is frozen. This is a medical emergency requiring hospital treatment.

First aid for frostbite:

Step 1. Get the person to a warm environment. Treat any hypothermia first, as it is more immediately life-threatening.

Step 2. Remove jewellery, rings, watches, and tight clothing from the affected area — swelling will occur during rewarming.

Step 3. For frostnip, warm the area gently using body heat — cup it with warm hands, place cold fingers in armpits, or breathe warm air on the area.

Step 4. For deeper frostbite, rewarm in warm (not hot) water at 37–40°C (99–104°F). This is water that feels comfortably warm to an unaffected body part. Too hot causes burns on numb skin.

Step 5. Do NOT rub frostbitten tissue. Ice crystals inside the tissue can tear cells when rubbed. Gentle handling only.

Step 6. Do NOT rewarm frostbitten tissue if there is any risk of it refreezing before reaching medical care. Refreezing after thawing causes dramatically worse tissue damage than the original frostbite. It is better to keep the tissue frozen than to thaw and refreeze.

Step 7. Once rewarmed, cover the area loosely with sterile or clean bandages. Separate frostbitten fingers and toes with cloth to prevent them sticking together.

Step 8. Get the person to medical care. Deep frostbite requires hospital treatment.

SECTION 4: PROTECTING YOURSELF IN A COLD WAVE

Layering: Wear three layers. A moisture-wicking base layer (avoid cotton — it stays wet). An insulating middle layer (wool or fleece). A windproof and waterproof outer layer. Staying dry is as important as staying warm.

Cover extremities: Fingers, toes, ears, and nose lose heat fastest. Gloves, thick socks, a hat covering the ears, and a scarf covering the nose and mouth are essential.

Stay dry: Wet clothing kills in the cold. Change out of wet clothing immediately, even if it means being briefly exposed to cold.

Avoid alcohol: Alcohol feels warming but actually causes the body to lose heat faster by dilating blood vessels. It also impairs judgement and reduces awareness of dangerous cold.

Eat regularly: Your body uses food to generate heat. In severe cold, the body burns calories much faster. Eat warm, calorie-rich food frequently.

Shelter: Wind dramatically increases heat loss. Even moderate cold with strong wind can be as dangerous as severe cold with no wind. Always seek shelter from wind as your first priority.

CHECK ON VULNERABLE PEOPLE DAILY during a cold wave. Elderly people living alone and infants are the most common victims of cold wave mortality. A brief daily visit or phone call can save a life.